Md Akib Haider

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Research Interests: Machine Learning, Human-Centric AI, Computer Vision

EDUCATION

- **Islamic University of Technology (IUT)** Aug 2022 – Present
  B.Sc. in Computer Science and Engineering
  - **CGPA**: 3.89/4.0 [upto 7th Semester].
  - **Supervisors:** Dr. Rafsanjany Kushol, Dr. Md. Hasanul Kabir.
- **Notre Dame College (NDC)** 2019–21
  Higher Secondary School Certificate (HSC), Science
  - **GPA**: 5.0/5.0 **(Government Merit Scholarship)**.

RESEARCH EXPERIENCE

- **LoRA-MME: Multi-Model Ensemble of LoRA-Tuned Encoders for Code Comment Classification**
  Published in ICSE-NLBSE | Paper 2026
  - We proposed a weighted stacking ensemble learning method while fine-tuning four complementary code encoders to solve multi-label classification across Java, Python, and Pharo in total 18 classes.
  - Our LoRA-based PEFT approach and per-category threshold optimization reduces total trainable parameters and achieves better classification performance over baseline.
- **Probabilistic Feature Imputation and Uncertainty-Aware Multimodal Federated Aggregation**
  Published in MIDL | Paper 2026
  - We proposed a probabilistic federated learning framework that replaces point estimates with calibrated distributions for uncertainty-aware local fusion and weighted aggregation across missing modalities.
  - Our "learning to distrust" paradigm significantly outperforms deterministic baselines on chest X-ray classification, generalized for any data-scarce federated environments.
- **Tri-Level Calibration for Training-Free Few-Shot Class-Incremental Learning with Label-Free Query Prototype Rectification** Paper | Under Review
  - We propose Tri-Level Calibration (TRIC) in FSCIL, semantic calibration within the language branch, inter-encoder calibration that fuses CLIP and DINO visual evidence on a common score axis, and inter-modal calibration between resulting visual and semantic evidence for fully inference based classification.
  - Combining with our Query Prototype Rectification (QPR), a fully label-free module that uses the unlabeled query batch to refine only the noisy novel visual prototypes, TRIC consistently improves all session accuracy over strong SOTA baselines, with positive gains in every seed, without any additional annotation.
- **PTPC: Two-Stage Patch-Token Prototype Calibration for Few-Shot Object Detection** Paper | Under Review
  - A two-stage Patch-Token Prototype Calibration, encoding query images once via a frozen foundation-model encoder for fully inference-based confidence recalibration of any CNN-based FSOD detector.
  - Our Base-Confusability Guard, suppresses detections whose crop-level features are closer to a cached base-class prototype than to any novel prototype, reached mean VOC AP50 of 62.7 paired with MFDC, surpassing the strongest CNN-based baseline by 2.0 points; halves Expected Calibration Error and Brier score.
- **Semi-Supervised Frequency-Aware GAN for Ultra Low to High Field MRI Translation** | Work in Progress
  - We presented a semi-supervised frequency-aware attention-guided framework for synthesizing 3T MRI from 64mT ultra-low-field scans, addressing paired data scarcity through adversarial refinement.
  - Our method preserves anatomical fidelity while restoring high-frequency 3T textures, outperforming recent baselines across T1, T2, and FLAIR contrasts with effective perception-distortion tradeoff.

WORK EXPERIENCE

- **Undergraduate Research Intern** Jun 2026 – Present
  University of Illinois Urbana Champaign, SRSE program, `Sharifa Sultana's Lab`
- **App Design and Development, Remote** Feb 2025 – Apr 2025
  SwadeshBazar, Chelmsford, UK
  - Our team built a cross-platform application (web and android) for a grocery delivery and shopping service start-up using Flutter, NextJs and Rave.
- **Industrial Training (Intern) on Mobile Application Development** Sep 2025 – Oct 2025
  Battery Low Interactive Ltd, Dhaka, Bangladesh
  - Learnt Flutter framework, Dart programming, State Management, API Integration, UI/UX design, App Testing and Deployment; collaborated on **Animeverse** (see Projects).

GRANTS AND FELLOWSHIPS

- **ACM SIGHPC Travel Grant**
  - HPC Immersion Program, Attended SC 2025 in St. Louis, Missouri, USA ($3000). 2025
  - Lightning Talk on "Data, Model, and Tensor Parallelism for efficient MRI Translation".

ACHIEVEMENTS

- **NLBSE Tool Competition** winner, co-located workshop with ICSE-26. 2026
- **Programming Contest**
  - **26th** (among 110 teams), IUT 12th National ICT Fest Programming Contest. 2026
  - **80th** (among 162 teams), Uttara University Inter-University Programming Contest. 2025
  - **6th**, Intra-IUT Junior Programming Contest. 2024
  - **52nd** (among 115 teams), IUT 11th National ICT Fest Programming Contest. 2024
  - **102nd** (among 317 teams), ICPC Asia Dhaka Regional Onsite Contest. 2024
  - Personal best: Country **rank 37**, Global **rank 1536** in Codeforces | profile. 2022-26
- **Hackathon**
  - **Champion** in Coderush 2.0, IUT-ICT Fest Hackathon | code 2025
- **Software Sub-Team Member, Project Altair** at IUT Robotics Society. 2023–24
  - **Global 6th place** and **Best Science Team award** in International Rover Challenge (IRC).

Games & Sports

- **Captain,** Inter University Chess Team of IUT | *FIDE Blitz Rating: 1553*. 2025–26
- All Bangladesh **rank 49** at 37th National Sub-Junior Chess Championship. 2018

Literature & Creative Work

- **Delegate (Writing)** at 6th International Children Film Festival. 2013
- **UNICEF-Meena Media Award** winner for creative work in print media. 2012–13

PROJECTS

- **Better_SIS, University Management System** | *Dart, C++, Swift, Flutter* | code 2025
  - Student Information System of Islamic University of Technology (IUT).
  - Academic Activity and Internal Service Management System for students, teachers and staff.
- **Student Driven Market Place** | *Flutter, Gemini 2.5, Open Street Map* | code 2025
  - Product selling and purchasing platform under virtually created institutional marketplace.
- **Animeverse** | *Flutter, Lmstudio, Dart, Docker, Jikan Api, Rive* | code 2025
  - Anime Streaming Platform: agentic features, suggestions, download, local share, social networking.
- **Xchange, File Sharing System** | *Socket Programming, Java* | code 2024
  - Secure, fast and efficient peer-to-peer file transfer on same local network, without internet.
- **E-Commerce** | *Dart, C++, Rive, Flutter, Docker* | code 2024
  - Android Working Prototype for Bazar Sodai.

LEADERSHIP & EXTRA CURRICULAR ACTIVITIES

- **Instructor, IUT ACM Community** Aug 2024 – Present
  - Mentor for sophomore Competitive Programming aspirants.
  - Organized and conducted weekly classes.
- **Organizer, Intra-IUT Junior Programming Contest** 2025
  - Event head at *Coderush 2.0*, organized by IUT Computer Society.
- **Vice President, Notre Dame English Club** 2019–22
  - Organized *6th National English Carnival*, Literature & Pop-Culture Quiz: *Litbuzz 3.0*.
- **Problem Setting**
  - Intra-IUT Junior Programming Contest | problemset 2025
  - Abid Hasan Intra University Programming Contest | problemset 2025
- **Anchoring and Hosting**
  - *37th Annual Games and Sports Day & Prize Giving Ceremony* at IUT. 2025
  - *Annual Cultural & Prize Giving Ceremony* at Ideal School & College. 2019
  - *11th & 12th IELC workshop* on 'English Olympiad', 'Extempore Speech & Turncoat'. 2018–19

Sports: Passionate about playing table tennis and chess.

REFERENCES

- **Academic Reference**: Dr. Rafsanjany Kushol, Postdoctoral Scholars, Faculty of Medicine and Dentistry - Medicine Dept, University of Alberta, Email: kushol@ualberta.ca, Phone: (+1) (587) 594-8658
- **Academic Reference**: Dr. Md. Hasanul Kabir, Professor, Department of CSE, Islamic University of Technology(IUT), Email: hasanul@iut-dhaka.edu, Phone: (+88 02) 996691254 59